

WEEK-10

PART 1: ACTIVATED SLUDGE PROCESS (ASP)

Definition (VERY IMPORTANT)

- **Activated sludge** = sludge containing **high concentration of active aerobic microorganisms**

MCQ Trap:

- Aerobic (NOT anaerobic)
-

Principle of ASP

- Activated sludge + raw sewage + **large quantity of air**
 - Aeration time: **4–8 hours**
 - Settled sludge in **secondary sedimentation tank (SST)** = activated sludge
 - Part of sludge is:
 - **Recycled** → **aeration tank**
 - **Excess** → **sludge treatment**
-

Flow Diagram (Sequence – VERY IMPORTANT)

PST → Aeration Tank → SST → Chlorination

Side processes:

- Return sludge → aeration tank
- Excess sludge → thickener → digestion

MCQ Trap:

- Recycling is from **secondary clarifier**, not primary
-

Components of ASP

1. **Aeration Tank**
 - Air supplied via compressor
 - Microbial degradation
 2. **Final Clarifier (Secondary Sedimentation Tank)**
 - Settling of biomass
 3. **Recycle Pump**
 - Returns sludge
-

Detention Times (VERY IMPORTANT NUMBERS)

- **Aeration tank** → 4 to 8 hours
- **Secondary sedimentation tank** → 1.5 to 2 hours
- **Primary settling tank** → 1.5 to 2 hours

👉 **MCQ Trap:**

- Aeration time > settling time
-

Bulking of Sludge (Exam Favorite)

- Occurs in **sick conditions**
- Sludge becomes:
 - Moist
 - Swollen
 - Does NOT settle
- Carried with effluent

✓ **Remedies:**

1. Remove industrial waste
2. Chlorination
3. Increase aeration
4. Raise pH to **~8 (using lime)**

👉 **MCQ Trap:** Bulking = **poor settling**



ASP VARIABLES (VERY IMPORTANT FORMULAS)

1 Hydraulic Retention Time (HRT)

$$\text{HRT} = \frac{V}{Q}$$

- V = volume of aeration tank
- Q = flow rate

👉 Meaning: **Average time sewage stays in aeration tank**

2 Volumetric BOD Loading (v)

- BOD load per unit volume

👉 Units: kg BOD / m³-day

3 Food to Microorganism Ratio (F/M)

- Ratio of:
 - Food (BOD)
 - Microbial mass (MLSS)

👉 **VERY IMPORTANT:**

- Lower F/M → **Higher BOD removal**
-

4 Sludge Age / MCRT (θ_c)

- Time solids remain in system

👉 **MCQ Trap:**

- Also called:
 - Mean Cell Residence Time (MCRT)

5 Sludge Volume Index (SVI)

- Volume occupied by **1 g sludge after 30 min settling**

Formula:

- $SVI = (\text{Volume in ml}) / (\text{MLSS in g})$

👉 Typical:

- **50–150 ml/g**
-



NUMERICAL (IMPORTANT)

✓ Given:

- $Q = 1 \text{ MLD} = 41.66 \text{ m}^3/\text{hr}$
- $V = 200 \text{ m}^3$

✓ HRT:

$$HRT = \frac{V}{Q} = \frac{200}{41.66} = 4.8 \text{ hours}$$

👉 MCQ Answer: 4.8 hours

✓ MCRT Calculation:

$$MCRT = \frac{\text{Biomass in tank}}{\text{Biomass leaving tank}}$$

Given:

- $MLSS = 2000 \text{ mg/L}$
- $\text{Effluent solids} = 50 \text{ mg/L}$

👉 Result:

- **MCRT = 8 days**

👉 MCQ Trap: Units = days



TYPES OF SETTLING (VERY IMPORTANT)

1. **Type I** → Discrete settling
2. **Type II** → Flocculent settling
3. **Type III** → Hindered / Zone settling
4. **Type IV** → Compression settling

👉 MCQ Trap:

- Activated sludge → **Zone settling**
-



QUICK REVISION (HIGH-YIELD)

- Aeration time = **4–8 hrs**
 - $HRT = V/Q$
 - $MCRT = \text{days}$
 - $F/M \downarrow \rightarrow \text{efficiency} \uparrow$
 - Bulking = poor settling
 - SVI normal = **50–150**
 - ASP uses **aerobic microbes**
-



MCQ PRACTICE

Q1. Activated sludge process uses:

- A. Anaerobic bacteria
 - B. Aerobic bacteria ☒
 - C. Fungi
 - D. Algae
-

Q2. HRT depends on:

- A. Temperature
 - B. Volume & flow rate ☒
 - C. pH
 - D. Oxygen
-

Q3. Sludge bulking causes:

- A. Better settling
 - B. No effect
 - C. Poor settling & suspension ☒
 - D. Faster digestion
-

Q4. F/M ratio decrease leads to:

- A. Lower treatment
 - B. Higher BOD removal ☒
 - C. No change
 - D. Toxicity
-

Q5. Typical SVI value:

- A. 10–20
- B. 50–150 ☒
- C. 200–300
- D. 500

PART 2: SOLID WASTE MANAGEMENT (SWM)

WHY SOLID WASTE MANAGEMENT IS IMPORTANT (Conceptual MCQ)

- Prevent:
 - Land pollution
 - Water pollution
 - Air pollution (burning waste)
- Avoid:
 - Disease spread
 - Fire hazards
 - Environmental degradation

Case Study Trap:

- **Ghaziapur landfill (Delhi)**
 - Opened: **1984**
 - Capacity exceeded: **2002**
-

BASIC METHODS OF WASTE DISPOSAL

1. **Dumping**
 - Low-lying areas
 - Uncontrolled
2. **Burning**
 - Reduces volume
 - Causes air pollution
3. **Recycling**
 - Recovery of useful materials
4. **Waste Minimization**
 - Reduce resource & energy use

MCQ Trap:

- Burning ≠ sustainable solution



HISTORY (VERY IMPORTANT MCQs)

Bubonic Plague

- Caused by: **Yersinia pestis**
- Spread via: **rodents (rats)**
- Deaths: **~50% population in 14th century Europe**

👉 Diseases:

- Leptospirosis
- Salmonellosis
- Toxoplasmosis

👉 MCQ Trap:

- Poor waste → disease spread
-



PRESENT DAY: 3Rs (VERY IMPORTANT)

1. **Reduce**
 - Use less
2. **Reuse**
 - Use again
3. **Recycle**

👉 Hierarchy (MOST IMPORTANT ORDER):

1. Reduce
2. Reuse
3. Recycle
4. Energy recovery
5. Disposal

👉 MCQ Trap:

- Disposal = **least preferred**



LIFE CYCLE ASSESSMENT (LCA)

- Evaluates **environmental impact from cradle-to-grave**

👉 MCQ Trap:

- LCA = whole life cycle, not single stage
-



LANDFILLS (VERY IMPORTANT)

Problems:

- Land usage
- Water contamination (leachate)
- **Methane (CH₄)** production
- Odor issues
- Fire risk

👉 Important Facts:

- Methane = **25× more powerful than CO₂**
- CH₄ can be used as **energy source**

👉 MCQ Trap:

- Landfills are engineered (modern ones)
-



ALTERNATIVES TO LANDFILLS

1. Incineration

✓ Pros:

- Reduces volume
- Energy recovery

✗ Cons:

- Air pollution
 - Some waste not burnable
 - Expensive
-

2. Resource Recovery

- Compost
- Refuse-derived fuel

👉 MCQ Trap:

- Residue still needs disposal
-

DEFINITION OF SWM (IMPORTANT)

- Comprehensive system including:
 - Generation
 - Storage
 - Collection
 - Transport
 - Processing
 - Disposal
-

MULTIDISCIPLINARY FIELD

SWM involves:

- Political science
- Economics
- Public health
- Engineering (small part!)

👉 MCQ Trap:

- Engineering ≠ whole system
-



6 ELEMENTS OF SWM SYSTEM (VERY IMPORTANT)

1 Waste Generation

- Some waste:
 - Can be reused
 - Can be converted (e.g., food → CH₄)
-

2 Handling, Separation, Storage (AT SOURCE)

- Segregation:
 - Blue, green, white bins
- Separate:
 - Plastics, paper, metals
- Avoid contamination

👉 MCQ Trap:

- Source segregation = BEST practice
-

3 Collection

- Most expensive stage
 - Includes transport to facility
-

4 Processing / Transformation

- Shredding
 - Compaction
 - Composting
 - Incineration
-

5 Transfer & Transport

- Modes:
 - Truck
 - Rail (**cheapest**)
 - Barge
-

6 Disposal

- Final stage
 - Landfills
 - Engineered for:
 - CH₄ recovery
 - Leachate control
-



FACTORS INCREASING MSW

- Population increase
- Lifestyle changes
- Income level

👉 Composition:

- Mostly **organic matter**
-

SELF PURIFICATION OF WATER (VERY IMPORTANT)

Forces:

Physical:

- Dilution
- Dispersion
- Sedimentation

- Sunlight

Chemical:

- Oxidation
- Reduction

👉 MCQ Trap:

- Sunlight = physical factor
-



DO SAG CURVE (SUPER IMPORTANT)

- After sewage discharge:
 - DO ↓
 - BOD ↑
- Then:
 - DO recovers downstream

👉 Key terms:

- **Critical point**
- **DO minimum**

👉 MCQ Trap:

- DO lowest where pollution highest
-



QUICK REVISION (EXAM BLAST)

- 3Rs → Reduce > Reuse > Recycle
 - Methane = **25× CO₂**
 - Collection = **most expensive**
 - Rail = **cheapest transport**
 - Segregation at source = **best**
 - Landfill → CH₄ + leachate
 - DO sag → oxygen drops then recovers
-

MCQ PRACTICE

Q1. Most preferred waste management option:

- A. Disposal
 - B. Recycling
 - C. Reduce ☒
 - D. Incineration
-

Q2. Methane is:

- A. Less harmful than CO₂
 - B. Equal
 - C. 25× stronger greenhouse gas ☒
 - D. Not harmful
-

Q3. Most expensive stage:

- A. Processing
 - B. Disposal
 - C. Collection ☒
 - D. Storage
-

Q4. Cheapest transport:

- A. Truck
 - B. Rail ☒
 - C. Air
 - D. Ship
-

Q5. DO sag curve shows:

- A. Increase in DO
- B. Constant DO
- C. Decrease then increase in DO ☒
- D. Only decrease

PART 3: ADVANCED SOLID WASTE MANAGEMENT (MCQ MASTER)



GLOBAL WASTE COMPOSITION (VERY IMPORTANT)

- Food & green waste $\approx$ 44% (highest)
- Paper/cardboard $\approx$ 17%
- Plastic $\approx$ 12%
- Glass, metal, rubber, wood $\rightarrow$ smaller %

 MCQ Trap:

- Organic waste = **major component globally**
-



FACTORS AFFECTING WASTE COMPOSITION

- Geographic location
- Seasons
- Collection frequency
- Population diversity
- Recycling practices
- Public attitude
- Legislation

 MCQ Trap:

- Waste composition is **NOT constant**
-



PROPERTIES OF SOLID WASTE (VERY IMPORTANT FORMULAS)

1 Moisture Content

Moisture Content = $\frac{W_w - W_d}{W_d}$ Moisture Content = $\frac{W_d}{W_w - W_d}$

- W_w = wet weight
- W_d = dry weight

👉 Drying temperature: **105°C**

2 Ash Content

- Burn at **575°C**
 - Residue = ash
-

3 Volatile Matter

- Heat at **950°C** for 7 min
-

4 Fixed Carbon

- = 100 - (moisture + ash + volatile matter)

👉 MCQ Trap:

- Fixed carbon is calculated indirectly
-



CALORIFIC VALUE (IMPORTANT)

Net Calorific Value (NCV / HHV formula)

$$HHV = 0.338C + 1.422(H - O/8) \quad HHV = 0.338C + 1.422(H - O/8)$$

- C, H, O = % composition

👉 Measured using:

- **Bomb calorimeter**
-



WASTE GENERATION RATE

- Small towns → ~100 g/person/day
- Medium → 300–400 g/day
- Large → 500 g/day

👉 General range:

- **0.3 – 0.6 kg/person/day**
-



GLOBAL WASTE TREATMENT (IMPORTANT)

- Landfill ≈ highest
- Recycling + composting growing
- Incineration moderate

👉 MCQ Trap:

- Landfill still dominant worldwide
-



FUNCTIONAL ELEMENTS (DETAILS)

Storage Types

- Type I: 50–20 L (with lid)
 - Type II: 50 L
 - Type III: 50–200 L
 - Type IV: MS bins (~4.5 m³)
 - Type V: Masonry bins
-

Collection Systems (VERY IMPORTANT)

1 Hauled Container System (HCS)

- Container itself transported

2 Stationary Container System (SCS)

- Waste collected, container stays

👉 MCQ Trap:

- HCS = container moves
 - SCS = container fixed
-



INTEGRATED WASTE MANAGEMENT

Includes:

- Source reduction
- Recycling
- Composting
- Waste-to-energy
- Landfilling

👉 Priority order same as hierarchy



SOURCE REDUCTION (VERY IMPORTANT)

- Reduce material use
- Increase product life
- Reduce toxicity
- Reuse materials

👉 Facts:

- Can reduce waste by **~50%**

👉 Packaging:

- **50% volume**
 - **1/3 weight**
-

RECYCLING

Pros:

- Saves resources
- Reduces mining
- Reduces environmental impact

Cons:

- Can cause pollution (if poorly managed)
- Needs stable market

👉 Methods:

- Curbside
 - Drop-off
 - Mail-back
-

COMPOSTING

- Biological decomposition
- Needs:
 - Organic matter
 - Water

- Oxygen

👉 Not suitable for:

- Meat, bones, fats, preserved wood
-

WASTE TO ENERGY (VERY IMPORTANT)

Types of Energy Recovery:

- Heat
 - Electricity
 - Cogeneration
-

Combustion Types:

1 Mass Burning

- Direct burning of waste
- Removes hazardous materials first
- Air supplied

👉 Capacity:

- ~3000 tons/day
-

2 RDF (Refuse Derived Fuel)

- Waste shredded
- Converted to pellets
- Used as fuel

👉 **MCQ Trap:**

- RDF = processed fuel



INCINERATION SYSTEM PARTS (Conceptual)

- Combustion chamber
 - Grates (moving/fixed)
 - Air supply
 - Ash handling system
 - Gas cleanup
-



ASH DISPOSAL

- Sent to **ash landfill**
-



WASTE-TO-ENERGY

Pros:

- Reduces volume
- Produces energy
- Ash usable in construction

Cons:

- Expensive
 - Air pollution
 - Toxic ash
 - Public concern
-



QUICK REVISION (FINAL BLAST)

- Organic waste = **highest (~44%)**
 - Moisture → 105°C
 - Ash → 575°C
 - Volatile → 950°C
 - Collection systems:
 - HCS = movable
 - SCS = fixed
 - Source reduction → **50% reduction possible**
 - Packaging → **50% volume**
 - Waste generation → **0.3–0.6 kg/day**
 - RDF = pellet fuel
 - Incineration → energy + pollution
-

FINAL MCQs

Q1. Moisture content is determined at:

- A. 50°C
 - B. 105°C ☒
 - C. 575°C
 - D. 950°C
-

Q2. Ash content determined at:

- A. 105°C
 - B. 575°C ☒
 - C. 950°C
 - D. Room temp
-

Q3. Largest component of waste:

- A. Plastic
 - B. Metal
 - C. Food waste ☒
 - D. Glass
-

Q4. HCS system:

- A. Fixed container
 - B. Movable container ☒
 - C. No container
 - D. Underground
-

Q5. RDF means:

- A. Raw waste
- B. Refuse Derived Fuel ☒
- C. Reduced disposal factor
- D. Recycled dry fraction

PART 4: LANDFILLS + LEACHATE (FINAL EXAM BOOST)

WHAT IS A LANDFILL?

- Developed in **early 20th century**
- Defined as:
 - 👉 Land where solid waste is deposited to **change surface elevation**

👉 MCQ Trap:

- Landfill $\neq$ random dumping
 - It is **controlled disposal**
-



KEY FACTS

- **50–70% of MSW** goes to landfill
 - Modern landfills:
 - Do NOT accept hazardous waste
 - Have:
 - Liners
 - Gas control
 - Leachate collection
 - Monitoring systems
-



LANDFILL PROBLEMS (VERY IMPORTANT)

- Disease spread
- Odor
- Fires
- Groundwater contamination
- Gas emissions

👉 Controlled by:

- Sanitary landfill techniques + proper design
-

SANITARY LANDFILL (VERY IMPORTANT)

Types:

- Excavated & filled
- Area method (build above ground)
- Combination

Operation:

- Cover soil daily
- Compact waste
- Control access

👉 **MCQ Trap:**

- Compaction reduces volume
-

LEACHATE (VERY IMPORTANT)

Definition:

- Liquid formed when water passes through waste

👉 Sources:

- Rainfall
- Groundwater

👉 Problems:

- Contaminates groundwater

- Contains toxic chemicals
-



LANDFILL GAS

- Mainly:
 - **Methane (CH₄)**
 - CO₂

👉 Produced by:

- **Anaerobic decomposition**

👉 MCQ Trap:

- Gas = anaerobic (not aerobic)
-



REGULATORY REQUIREMENTS

Location Restrictions:

- Airports
 - Wetlands
 - Fault lines
 - Unstable areas
 - Protected zones
-

LINERS (VERY IMPORTANT)

👉 Purpose:

- Prevent leachate leakage

Types:

1 Single Liner

- One layer:
 - Clay OR geomembrane
-

Composite Liner

- Clay + geomembrane together

 **BEST COMMON TYPE**

Double Liner

- Two liners + leak detection system

 **MOST SAFE**



STRUCTURE DETAILS (IMPORTANT VALUES)

- Drainage layer: ~2 ft
- HDPE geomembrane: ~60 mil
- Soil layer permeability:
 - $K \leq 10^{-7} \text{ cm/s}$

 **MCQ Trap:**

- Very low permeability = good liner
-



GEOMEMBRANE MATERIALS

- HDPE
- PVC
- VLDPE
- PP

👉 MCQ Trap:

- HDPE most common
-

LEACHATE COLLECTION SYSTEM (VERY IMPORTANT)

Working:

- Gravity drainage through pipes
- Pumping from low points

👉 Key:

- Liner slope = **2–8%**
-

Process:

1. Leachate collected
 2. Moves via pipes
 3. Sent to:
 - Treatment
 - Storage
-



GEONET & GEOTEXTILE

Geonet

- HDPE material
 - Used for **drainage**
-

Geotextile

- Textile material
- Functions:
 - Filtration
 - Separation
 - Allows water flow

👉 **MCQ Trap:**

- Geotextile ≠ waterproof (it allows flow)
-



QUICK REVISION (FINAL MEMORY)

- Landfill = controlled disposal
 - 50–70% waste → landfill
 - Leachate = liquid from waste
 - Gas = $\text{CH}_4 + \text{CO}_2$
 - Liners:
 - Single
 - Composite
 - Double (best)
 - Permeability:
 - 10^{-7} cm/s
 - Slope:
 - 2–8%
 - Geonet → drainage
 - Geotextile → filtration
-



FINAL MCQs

Q1. Leachate is:

- A. Gas
 - B. Solid
 - C. Liquid formed by water through waste ✓
 - D. Air
-

Q2. Most effective liner:

- A. Single
 - B. Composite
 - C. Double liner ✓
 - D. None
-

Q3. Landfill gas is produced by:

- A. Aerobic process
 - B. Anaerobic process ✓
 - C. Chemical reaction
 - D. Sunlight
-

Q4. Permeability of liner:

- A. High
 - B. Medium
 - C. Very low (10^{-7} cm/s) ✓
 - D. Zero
-

Q5. Geotextile is used for:

- A. Waterproofing
 - B. Filtration & separation ✓
 - C. Burning
 - D. Storage
-



YOU ARE FULLY PREPARED

You now have:

- ASP ✓
- SWM basics ✓
- Advanced SWM ✓
- Landfill + leachate ✓

PART 5: LANDFILL GAS + STABILIZATION + COMPOSTING

LEACHATE QUALITY (ADVANCED CONCEPT)

- Leachate composition:
 - Depends on **type of waste**
 - Depends on **phase of landfill stabilization**

👉 MCQ Trap:

- Same landfill → different leachate quality over time
-

LANDFILL STABILIZATION PHASES (VERY IMPORTANT)

Order (REMEMBER THIS SEQUENCE):

1. **Preliminary Aerobic Phase**
2. **Acid Formation Phase**
3. **Methane Formation Phase**
4. **Final Aerobic Phase**

👉 Shortcut:

👉 **A → Acid → Methane → Aerobic**

Phase Characteristics:

1 Preliminary Aerobic

- Oxygen present

- Initial decomposition
-

2 Acid Formation

- Anaerobic
- **Acids produced** → pH decreases
- BOD high

👉 MCQ Trap:

- Lowest pH in this phase
-

3 Methane Formation

- Anaerobic
 - $\text{CH}_4 + \text{CO}_2$ produced
 - **pH increases (stabilization)**
-

4 Final Aerobic

- Stabilized landfill
 - Low activity
-



GRAPH INTERPRETATION (VERY IMPORTANT)

- pH:
 - ↓ in acid phase
 - ↑ in methane phase
- **BOD/VFA:**
 - Peak in acid phase
 - Declines later

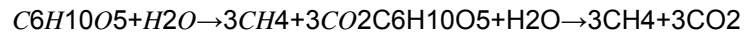
👉 MCQ Trap:

- Highest pollution → acid phase



LANDFILL GAS (VERY IMPORTANT)

Generation Reaction:



👉 MCQ Trap:

- Produces equal CH_4 and CO_2 (3:3 ratio)
-

Composition:

- Methane (CH_4)
 - CO_2
 - Water vapor
 - H_2S
 - NMOC
-

Why Important?

- Odor
 - Toxic
 - **Greenhouse gas**
 - Explosive
 - Energy source
-



GAS COLLECTION SYSTEM

Method:

- Vertical wells
- Installed after landfill filled

Types:

- **Passive wells** → open to atmosphere
 - **Active wells** → connected to extraction system
-

Working:

- Vacuum applied
- Gas extracted → pipes
- Sent to:
 - Flare
 - Energy recovery

👉 MCQ Trap:

- Gas extraction uses **pressure difference**
-

⚡ DRIVING FORCE FOR GAS FLOW

👉 Pressure gradient

- Gas moves from:
 - High pressure → low pressure
-



GAS GENERATION CURVE (IMPORTANT)

- Peaks after some years
- Then declines

👉 MCQ Trap:

- Not constant → follows curve
-



BIOLOGICAL METHODS

- Composting
 - Anaerobic digestion
-



COMPOSTING (VERY IMPORTANT)

Definition:

- Conversion of organic waste → **soil-like material**
-

Requirements:

- Organic matter
 - Water
 - Oxygen
-

NOT suitable for:

- Meat
 - Bones
 - Fat
 - Preserved wood
-



C:N RATIO (SUPER IMPORTANT)

- Ideal:
👉 20:1 to 25:1

👉 Effects:

- High C:N → slow process
- Low C:N → nitrogen loss

Examples:

- Yard waste → high C
 - Food waste → high N
-



TEMPERATURE (VERY IMPORTANT)

Phases:

1. **Mesophilic** → 30–35°C
2. **Thermophilic** → 50–60°C

👉 Important:

- 60°C for 24 hrs → kills pathogens

👉 >65°C → activity decreases



pH CONTROL

- Initial → **7–7.5**
- Drops to ~5 (acid formation)
- Then rises to ~8

👉 Final compost:

- pH ≈ 7

👉 MCQ Trap:

- Lime addition → NOT preferred
-



OTHER FACTORS

Moisture:

- Ideal → **50–60%**

Particle size:

- 25–75 mm

Mixing:

- First after 3 days
 - Then every 5 days
-



COMPOSTING TIME

- Mechanical → 7 days
- Windrow → 9–21 days
- Static → 30–40 days

👉 Curing:

- Additional **2–8 weeks**
-



COMPOSTING METHODS

1. Agitated

- Mixed regularly

2. Static

- Air supplied

3. Windrow

- Long piles
- Turned regularly

👉 Dimensions:

- Height $\approx$ 3 m
 - Width $\approx$ 5 m
-



FINAL ULTRA REVISION

- Stabilization:
 - Aerobic $\rightarrow$ Acid $\rightarrow$ Methane $\rightarrow$ Aerobic
 - pH $\downarrow$ then $\uparrow$
 - CH₄ production = methane phase
 - Gas extraction $\rightarrow$ wells + vacuum
 - C:N $\rightarrow$ **20–25**
 - Temp:
 - 30–35°C
 - 50–60°C
 - Moisture $\rightarrow$ **50–60%**
 - Compost curing $\rightarrow$ weeks
-



FINAL MCQs

Q1. Lowest pH occurs in:

- A. Aerobic
 - B. Acid phase ☒
 - C. Methane phase
 - D. Final phase
-

Q2. Ideal C:N ratio:

- A. 5:1
 - B. 10:1
 - C. 20–25:1 ☒
 - D. 50:1
-

Q3. Methane is produced in:

- A. Aerobic phase
 - B. Acid phase
 - C. Methane phase ☒
 - D. Final phase
-

Q4. Compost temperature for pathogen kill:

- A. 30°C
 - B. 40°C
 - C. 60°C ☒
 - D. 80°C
-

Q5. Driving force of landfill gas:

- A. Temperature
- B. Pressure gradient ☒
- C. Wind
- D. Gravity

PART 6: ANAEROBIC DIGESTION + GLOBAL WASTE (LAST BOOST)



ANAEROBIC DIGESTION (VERY IMPORTANT)



Definition

- Biological decomposition of organic waste **in absence of oxygen**



Output:

- **Biogas ($\text{CH}_4 + \text{CO}_2$)**
 - Digestate (fertilizer)
-



STAGES OF ANAEROBIC DIGESTION (VERY IMPORTANT ORDER)

1. **Hydrolysis**
2. **Acidogenesis**
3. **Acetogenesis**
4. **Methanogenesis**



Shortcut:

👉 **H → A → A → M**

Stage Details:

1 Hydrolysis

- Complex organics → simple sugars
-

2 Acidogenesis

- Produces:
 - Volatile fatty acids
 - Alcohols

3 Acetogenesis

- Converts acids → acetic acid
-

4 Methanogenesis

- Produces:
 - CH_4 (methane)
 - CO_2

👉 MCQ Trap:

- Methane ONLY in last stage
-

IDEAL CONDITIONS (VERY IMPORTANT NUMBERS)

- pH → 6.5 – 7.5
 - Alkalinity → 1000 – 5000 mg/L
 - Oxygen → ZERO
 - Nutrients → N & P required
-



Temperature

- Mesophilic → 30–38°C
- Thermophilic → 55–60°C

👉 MCQ Trap:

- Thermophilic = faster but sensitive
-



SYSTEM TYPES

1. Batch System

- Low CH₄ production
 - Large area
-

2. Two-stage System (IMPORTANT)

- Stage 1 → acidogens
- Stage 2 → methanogens

👉 Advantage:

- Better control
-



APPLICATIONS

- Wastewater treatment
 - Biogas production
 - Electricity
 - Fertilizer
 - Cooking gas
 - Vehicle fuel
-

COMPOST PLANT DESIGN (IMPORTANT MCQs)

Key Considerations:

- Storage for **≥ 2 days**
 - Uniform processing
 - Separation of inorganic waste
 - Pathogen-free compost
 - Odor control systems
-

Design Parameters:

- C:N ratio
 - Moisture
 - Windrow size
 - Compost storage → **3+ months**
-

GLOBAL WASTE (VERY IMPORTANT DATA)



Key Facts:

- Waste will increase **70% by 2050**
 - From:
 - **2.01 billion tons → 3.4 billion tons**
-



Fastest Growing Regions:

- Sub-Saharan Africa
- South Asia

👉 MCQ Trap:

- Not Europe or USA
-



Highest Waste Generation:

- East Asia & Pacific
-



WASTE VS INCOME (VERY IMPORTANT)

Collection Rates:

- High income → **96%**
 - Low income → **39%**
-

Waste Generation:

- Higher income → more waste

👉 MCQ Trap:

- Rich countries generate more waste per person
-



GDP vs Waste

- **Positive correlation**
👉 GDP ↑ → Waste ↑
-



DISPOSAL METHODS VS INCOME

- High income:
 - Recycling + incineration
 - Low income:
 - Open dumping
-



COMPOSITION VS INCOME

- Low income:

- More **organic waste**
 - High income:
 - More **plastic, paper**
-



SPECIAL WASTE

- Industrial waste → highest
 - Agricultural
 - Construction
 - Hazardous
 - Medical
 - E-waste
-



CLIMATE IMPACT (VERY IMPORTANT)

- Waste sector → ~5% global GHG emissions
 - Methane = **major contributor**
-

Reduction Methods:

- Recycling
 - Composting
 - Better collection
 - Gas capture
-



SOCIAL IMPACT

- **15 million people** depend on waste sector
 - Many:
 - Women
 - Children
 - Migrants
-



CIRCULAR ECONOMY (IMPORTANT)

- Replace:
 - Linear → (use → throw)
 - With:
 - Circular → reuse, recycle
-



FINAL ULTRA REVISION (COMPLETE COURSE)

- AD stages → H → A → A → M
 - pH → 6.5–7.5
 - Temp → 30–38 / 55–60°C
 - Methane → last stage
 - Waste ↑ with GDP
 - Collection:
 - High income → 96%
 - Low income → 39%
 - GHG → 5%
 - Waste ↑ 70% by 2050
-



FINAL MCQs

Q1. Methane is produced in:

- A. Hydrolysis
 - B. Acidogenesis
 - C. Acetogenesis
 - D. Methanogenesis ☒
-

Q2. Ideal pH for anaerobic digestion:

- A. 5
- B. 6.5–7.5 ☒
- C. 8.5
- D. 9

Q3. Fastest waste growth region:

- A. Europe
 - B. USA
 - C. Sub-Saharan Africa ☒
 - D. Japan
-

Q4. Waste sector contributes:

- A. 1%
 - B. 3%
 - C. 5% ☒
 - D. 10%
-

Q5. Collection rate in low-income countries:

- A. 20%
- B. 39% ☒
- C. 60%
- D. 90%